

10.10.2025

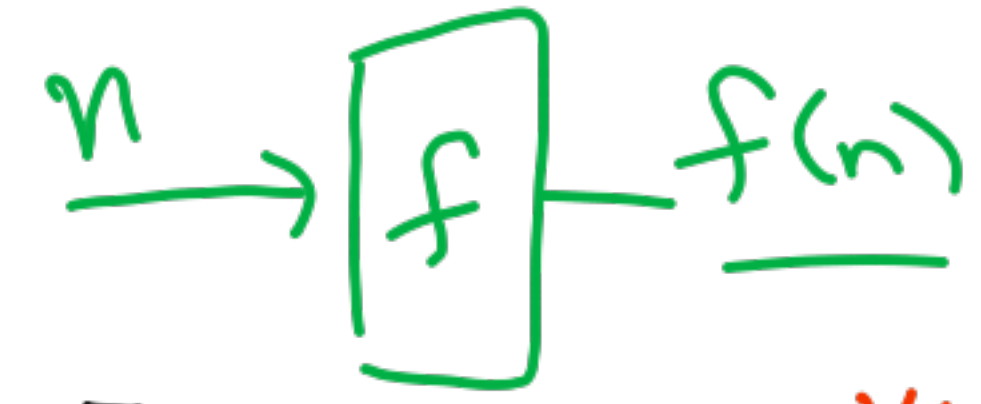
Quiz 2
15/10
11-12:30

① λ -calculus

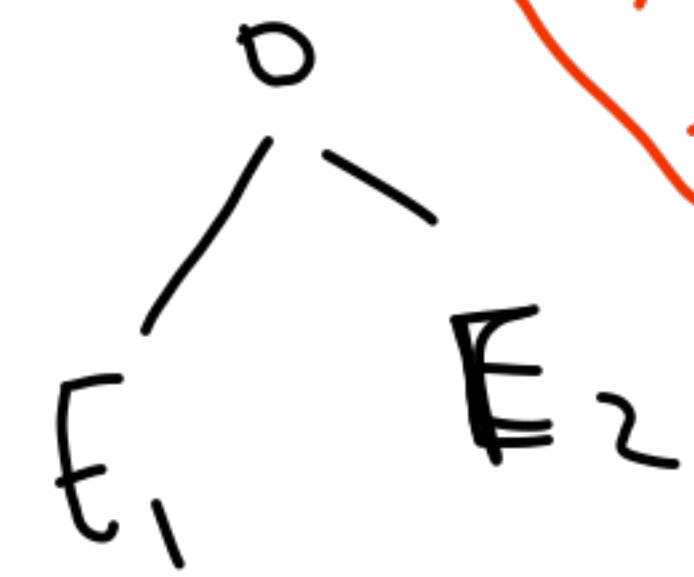
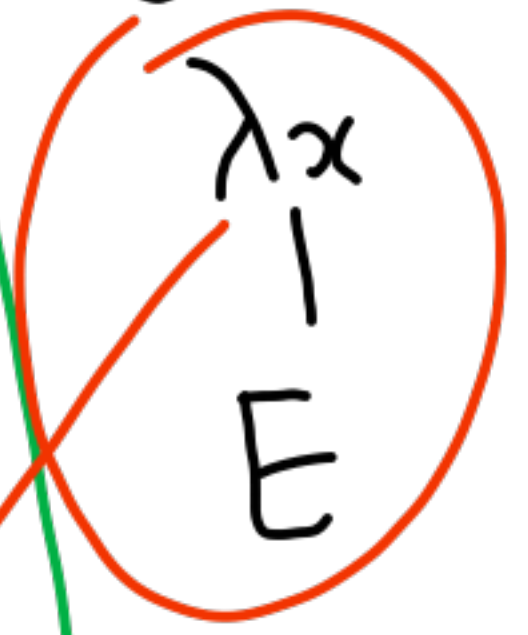
② Term Rewriting

③ Recursive Function Theory

Not in Quiz 2
↓
Quiz 3 &
I will post pointers on moodle



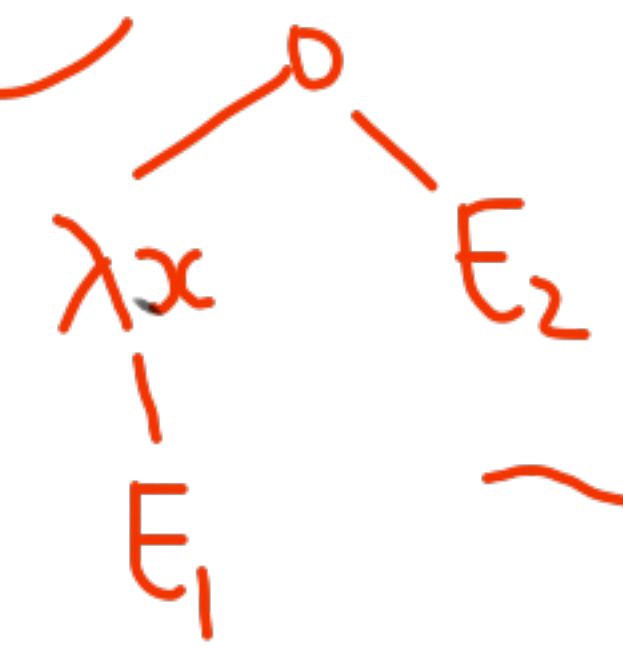
λ -Expression → Variables
 x, y, z
 f, m^n



Abstraction
formal $\approx$ parameter

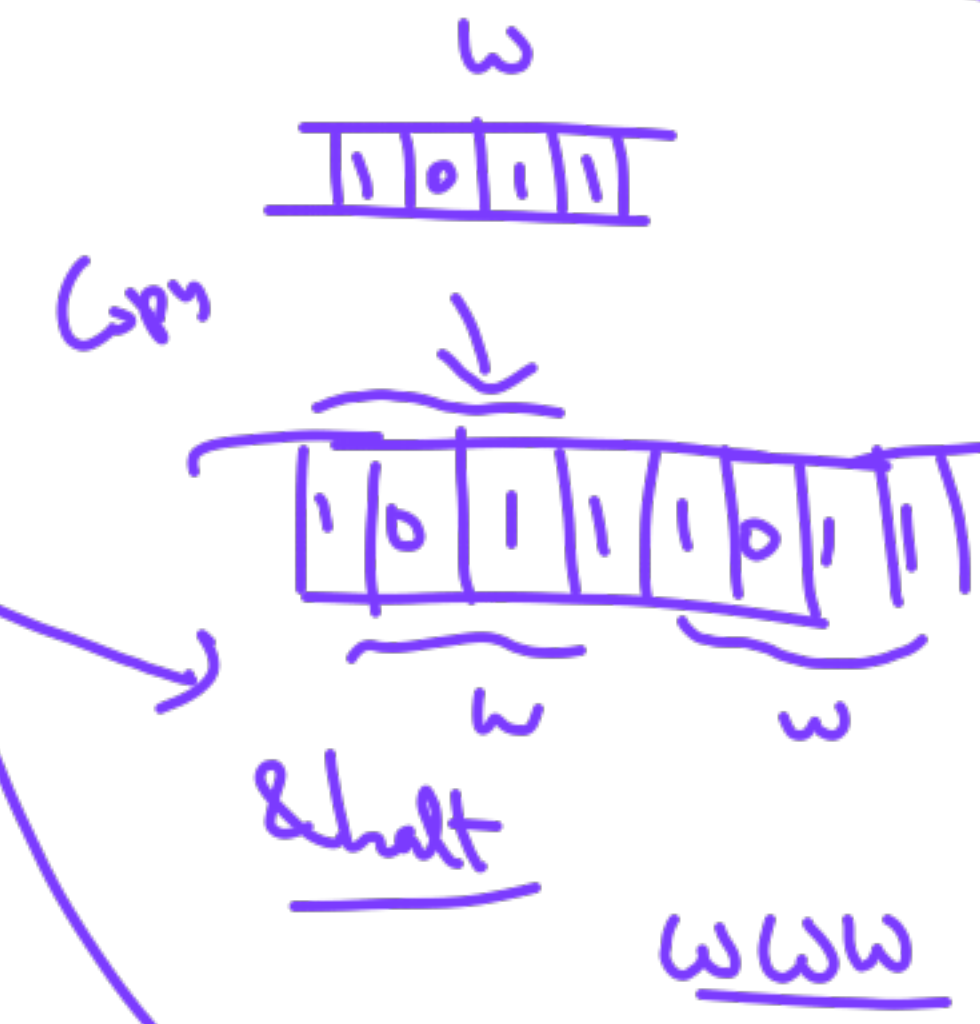
Application

β -redn



$\xrightarrow{\beta} E_1[x \rightarrow E_2]$

Type 0
TM practice



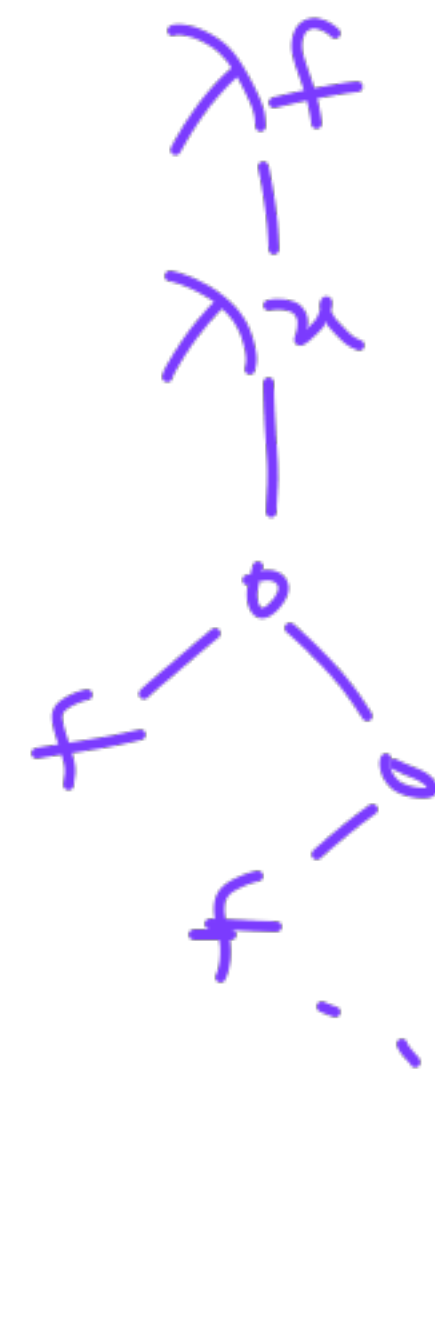
Church numerals

Numbers

abstraction → 0



23



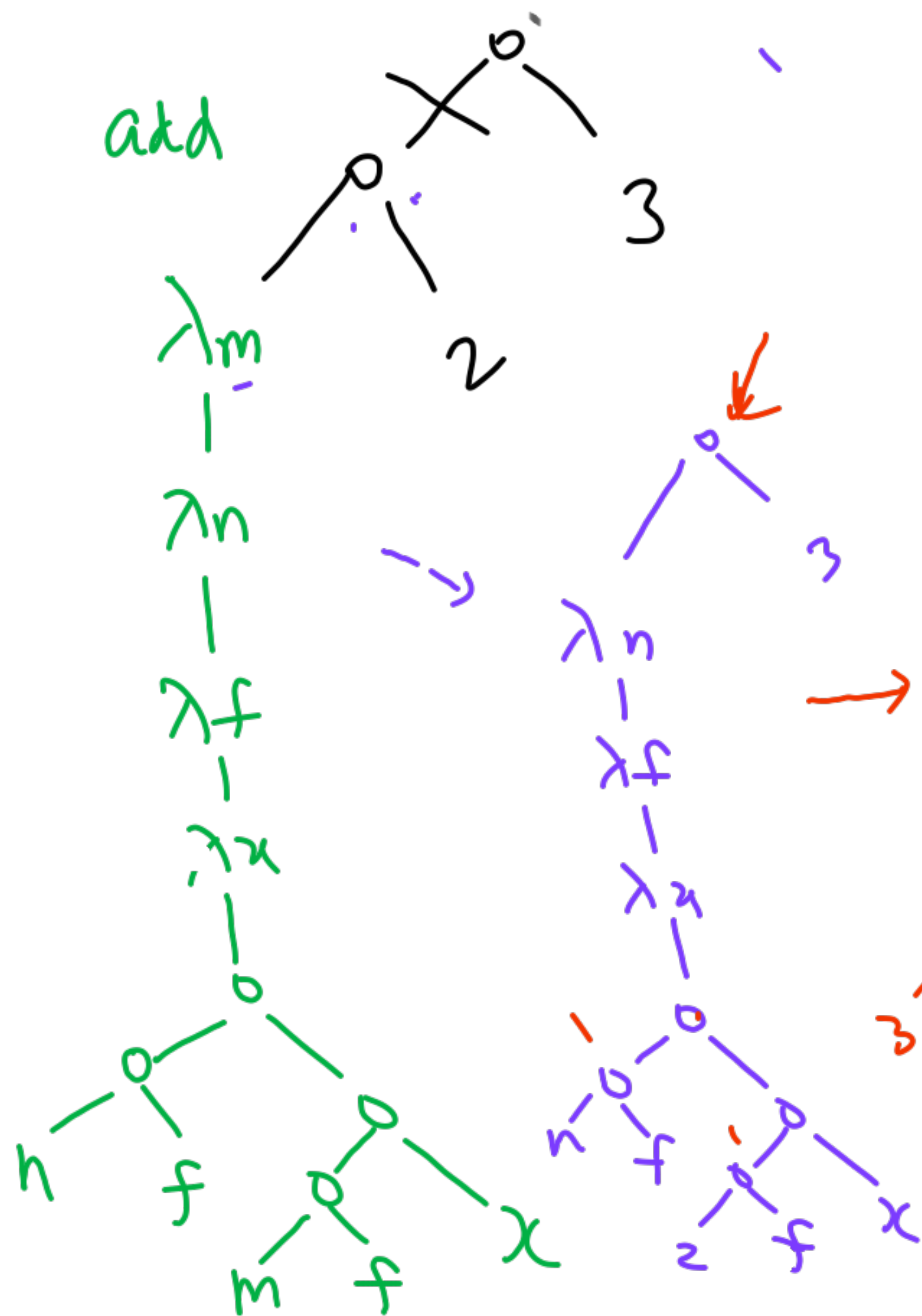
Goal
 Succ add mult exp



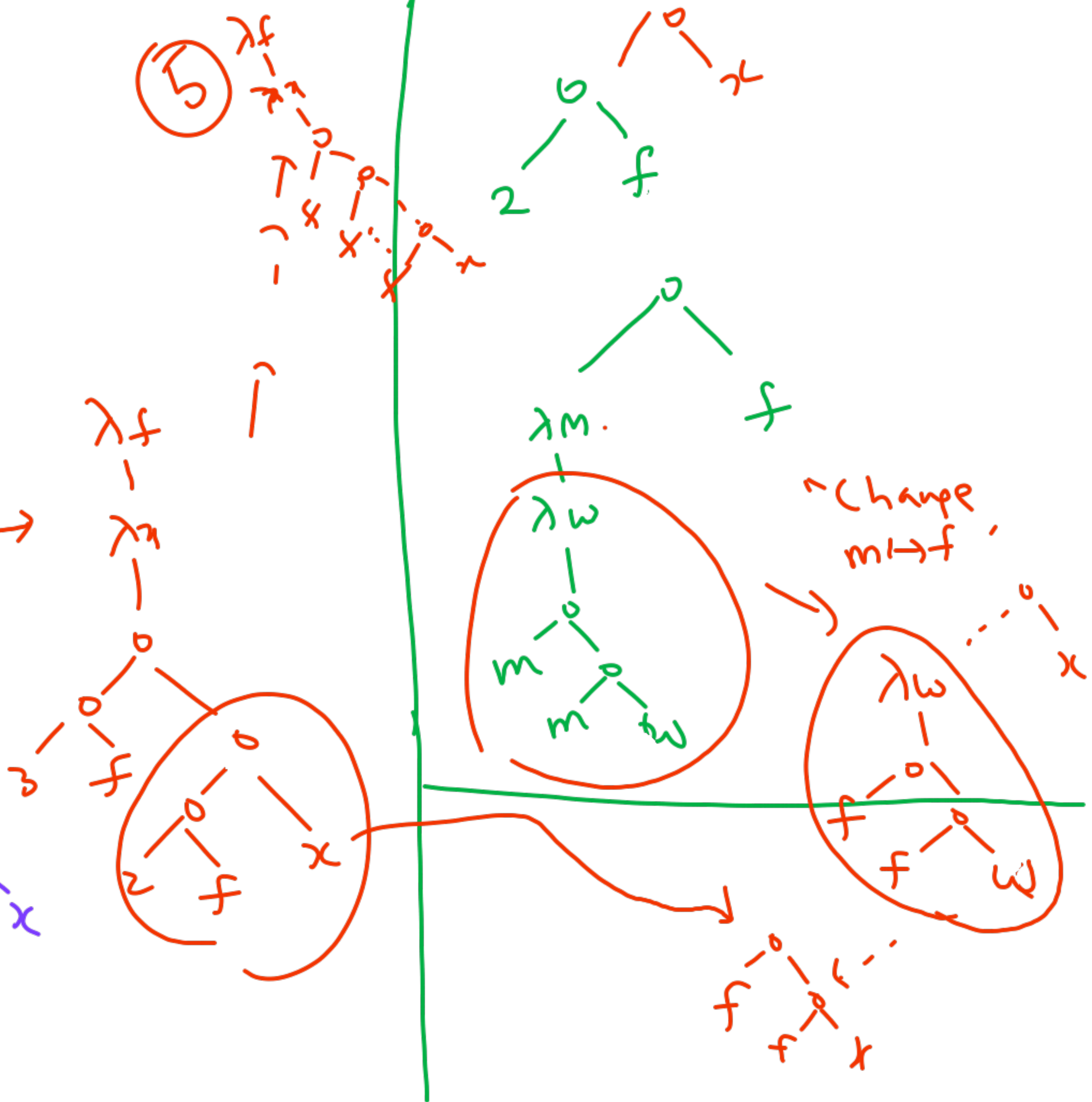
(mult)

12

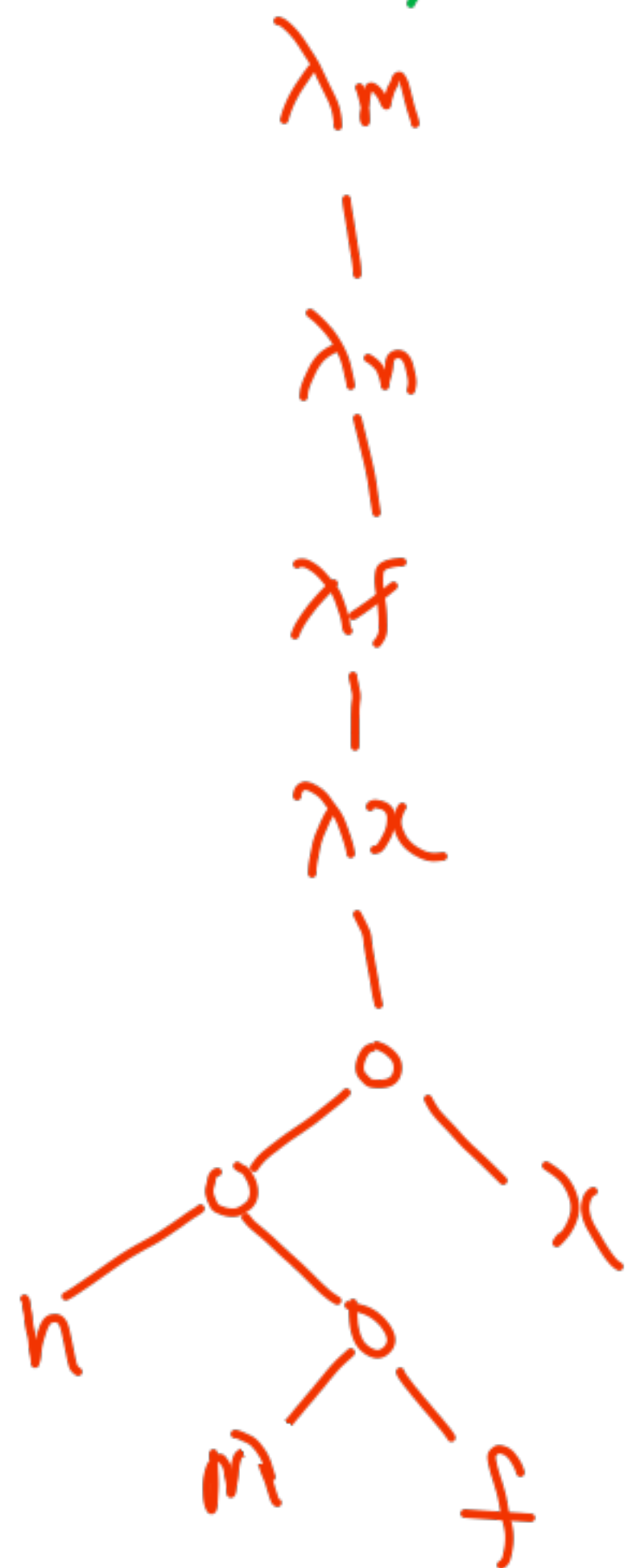
add



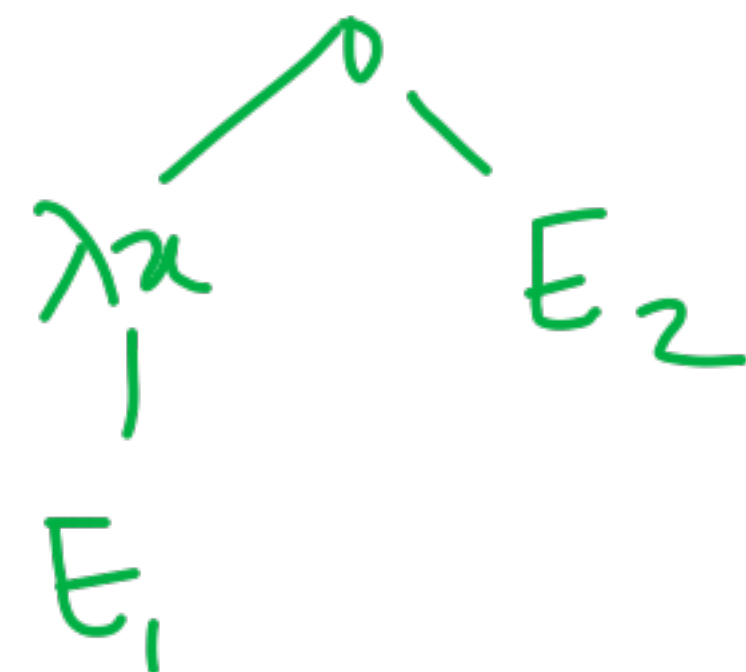
(5)



mult



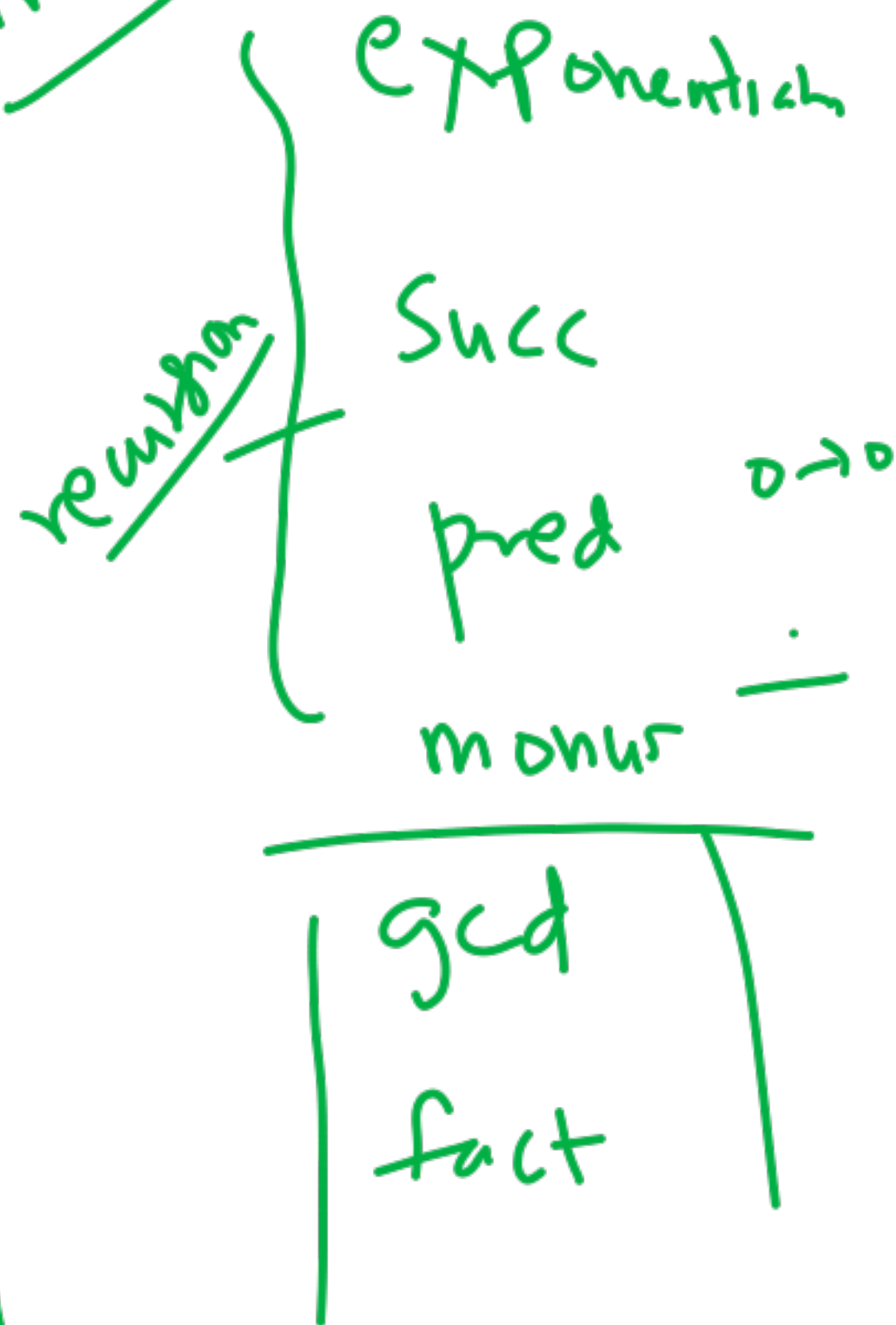
β-reduction



$E_1[x \mapsto E_2]$

"replace x by E_2 in E_1 "

Arithmetik



Boolean

False

True

λx
|
 λy
|
 y

λx
|
 λy
|
 x

and

λp
|
 λa

or $\Rightarrow$ not

Reduction

$Y \rightarrow E \rightarrow \beta \cdot E \rightarrow Y \rightarrow E$

1-step unrolling of reduction

Y is $\lambda g \lambda x \lambda a$
|
 λx
|
 g
|
 x
|
 x

λa
|
 g
|
 x
|
 x

if-then-else

test $\equiv$
then $\equiv$
else $\equiv$

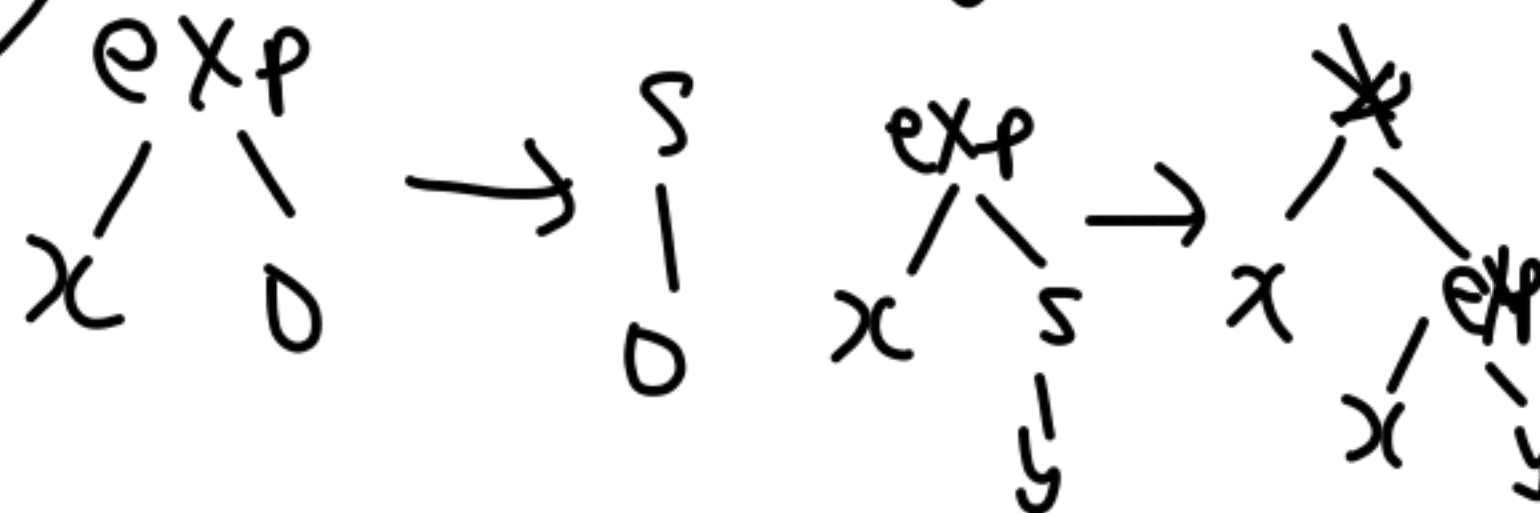
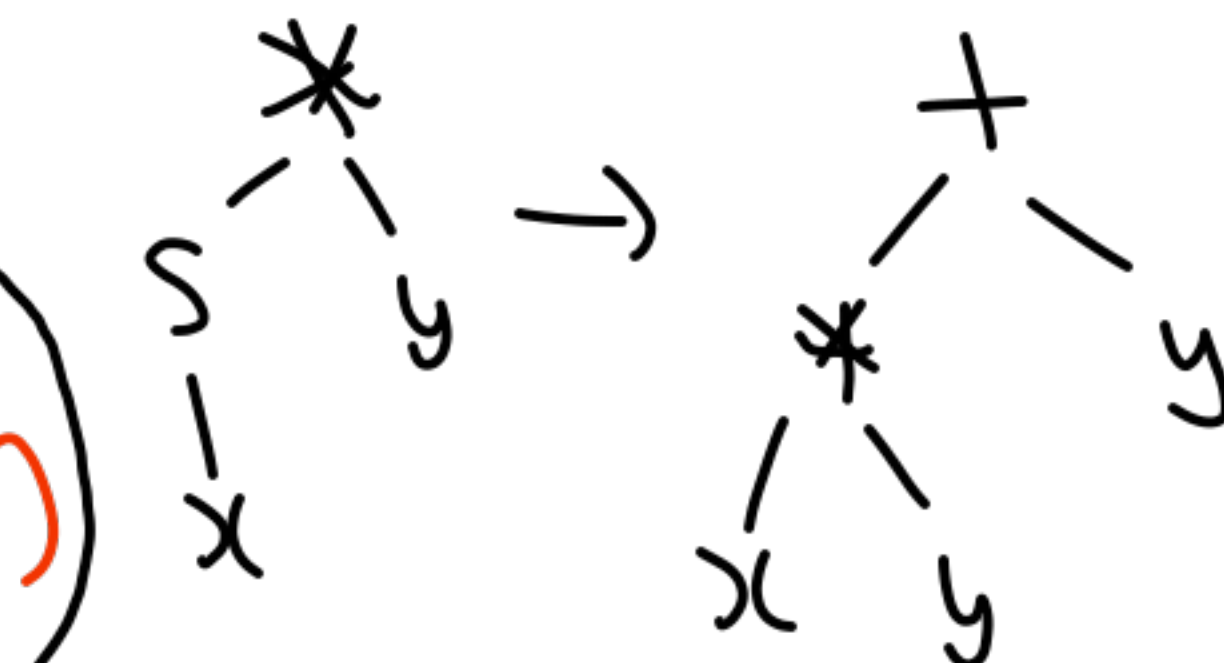
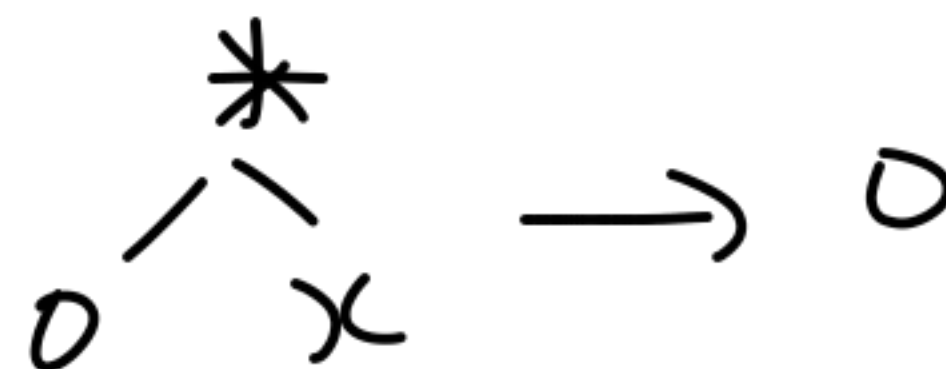
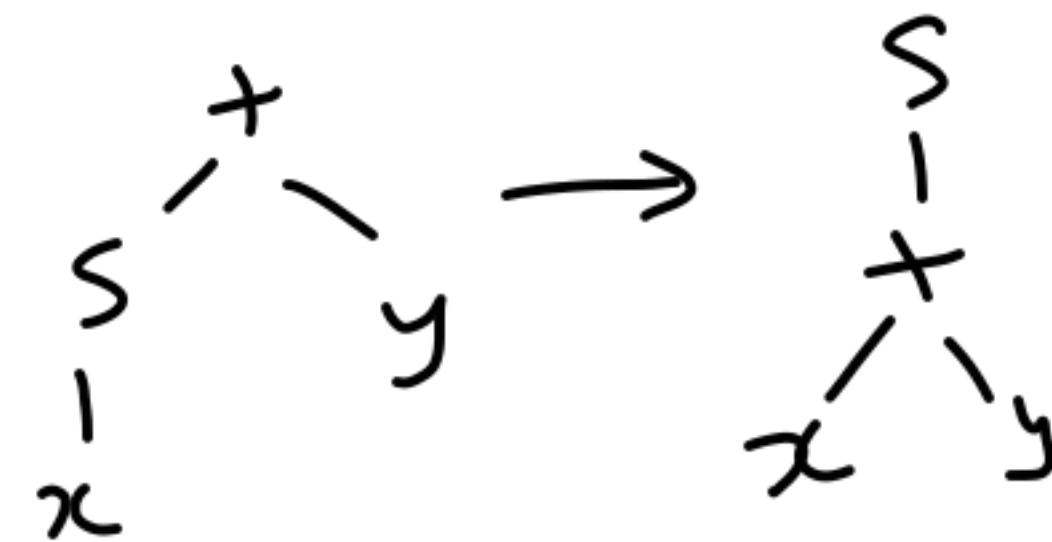
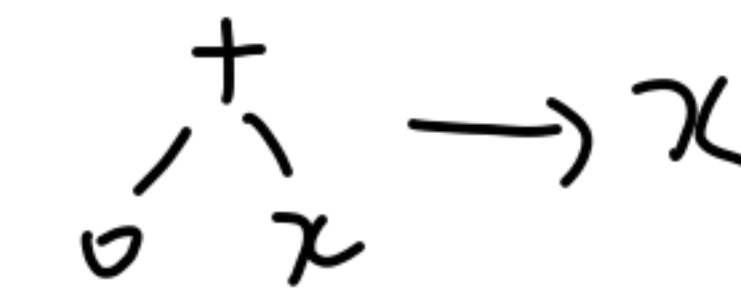
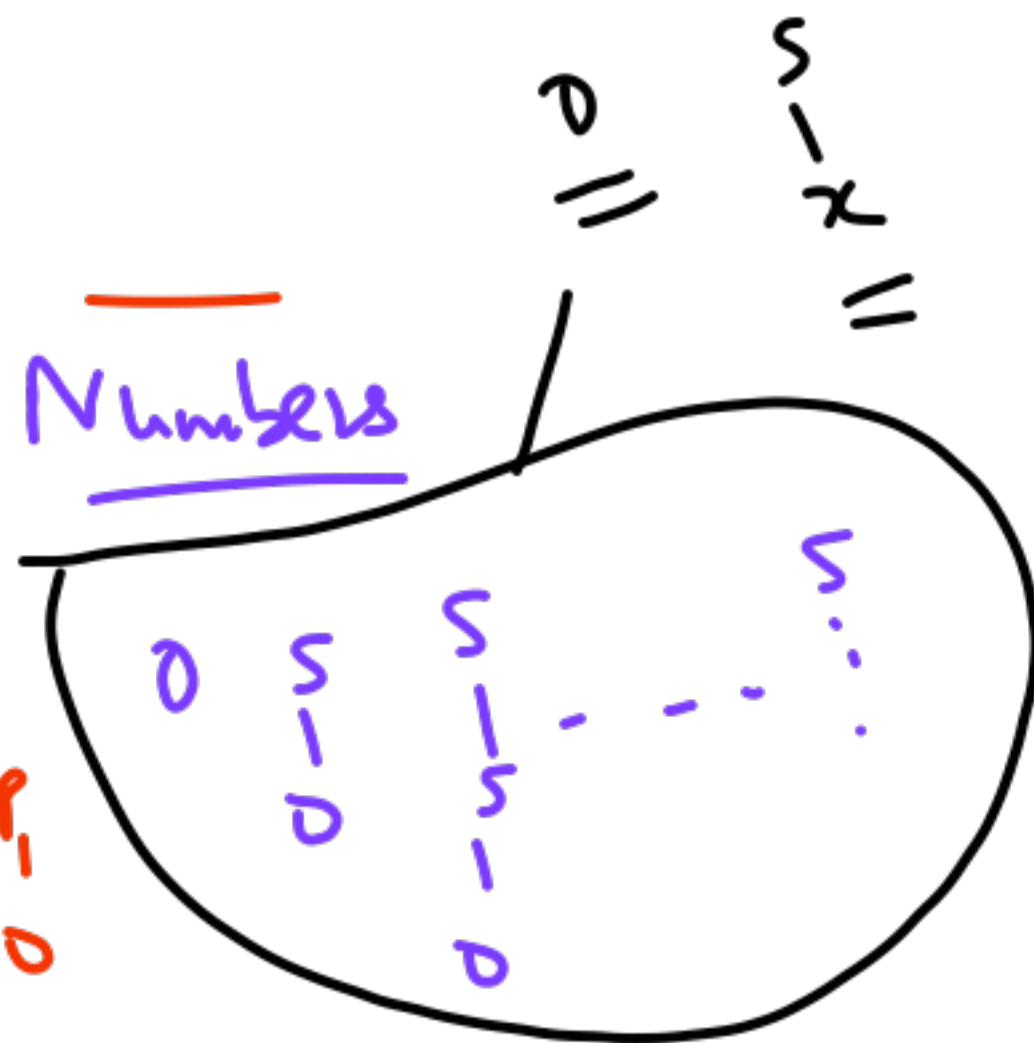
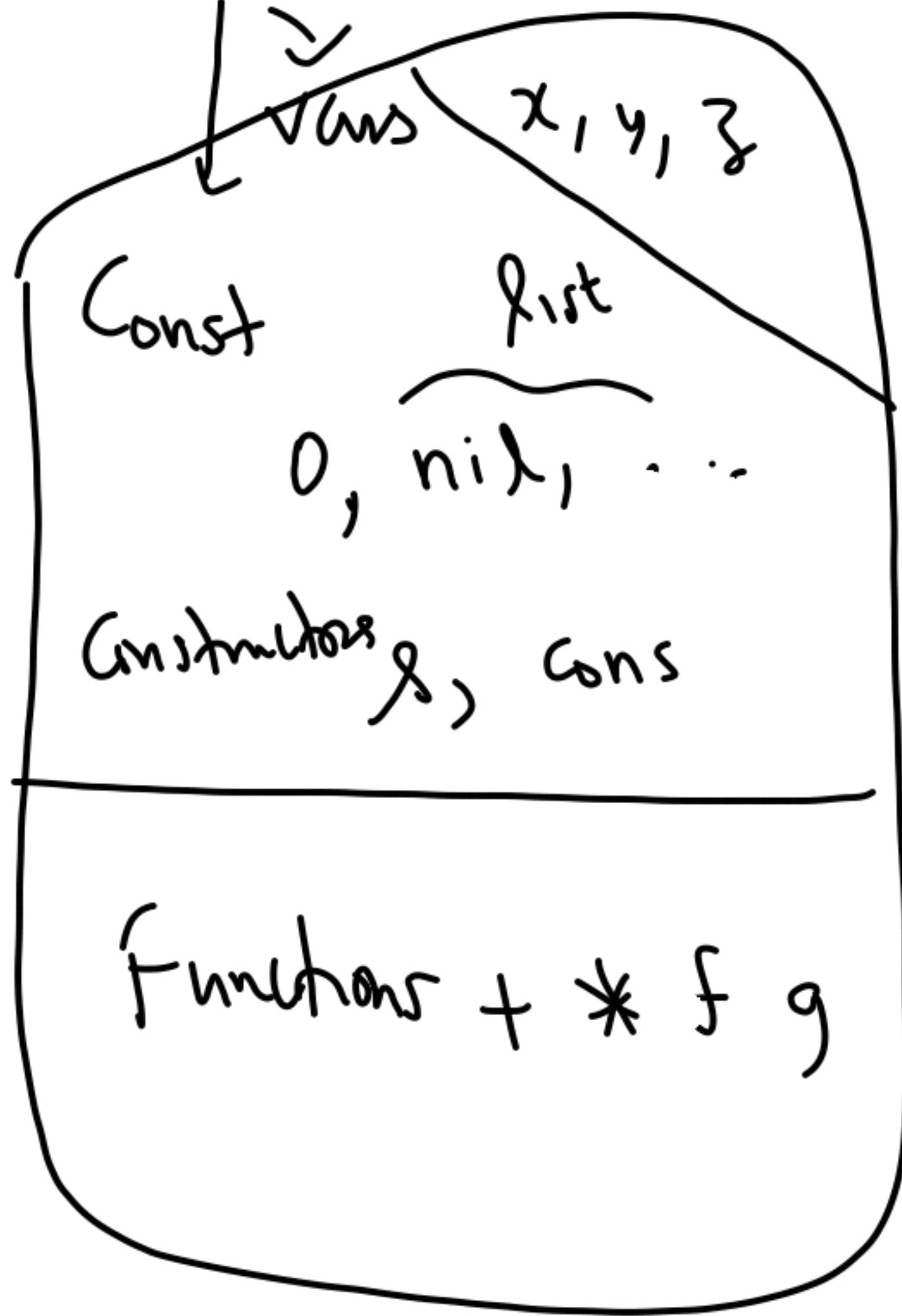
else $\equiv$

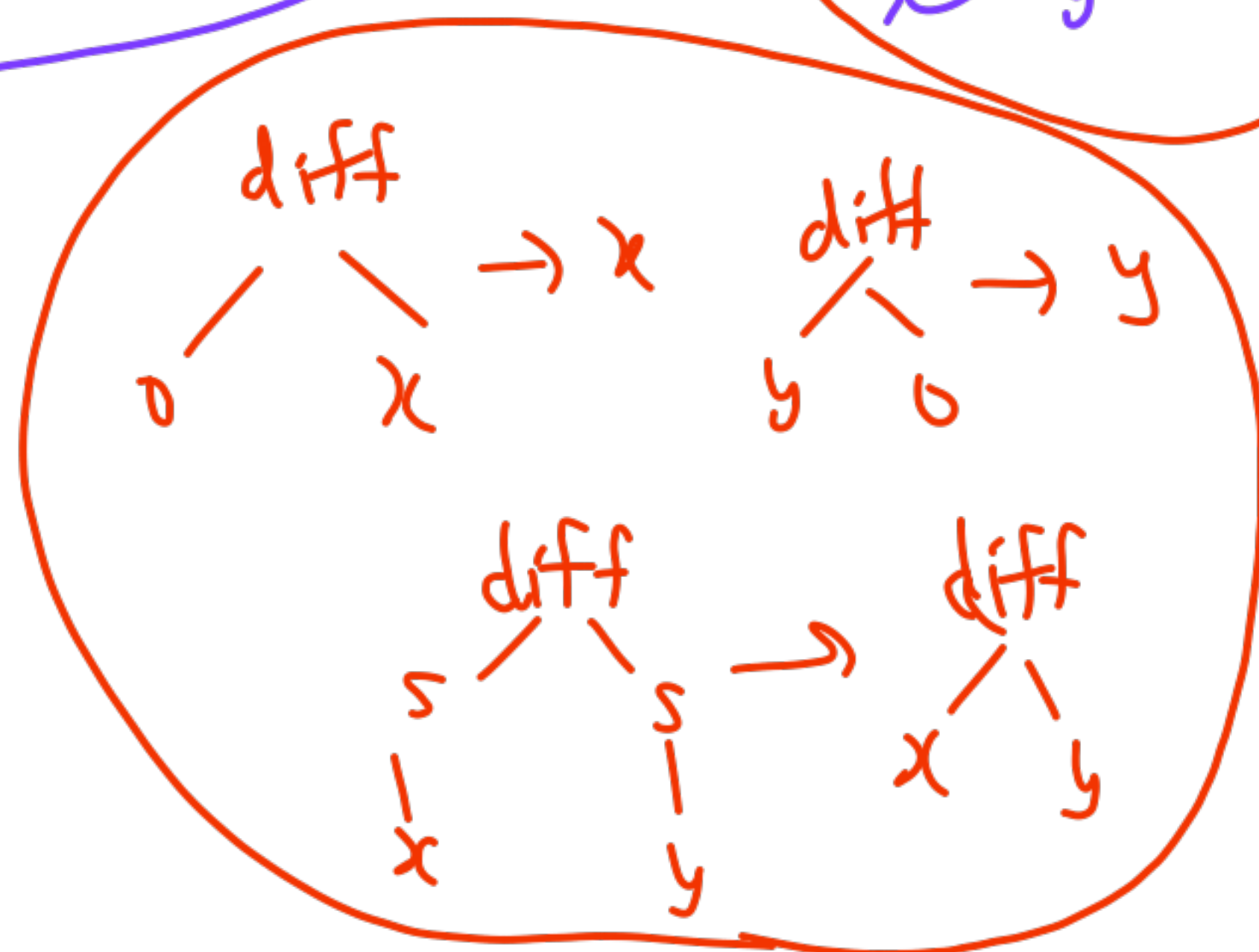
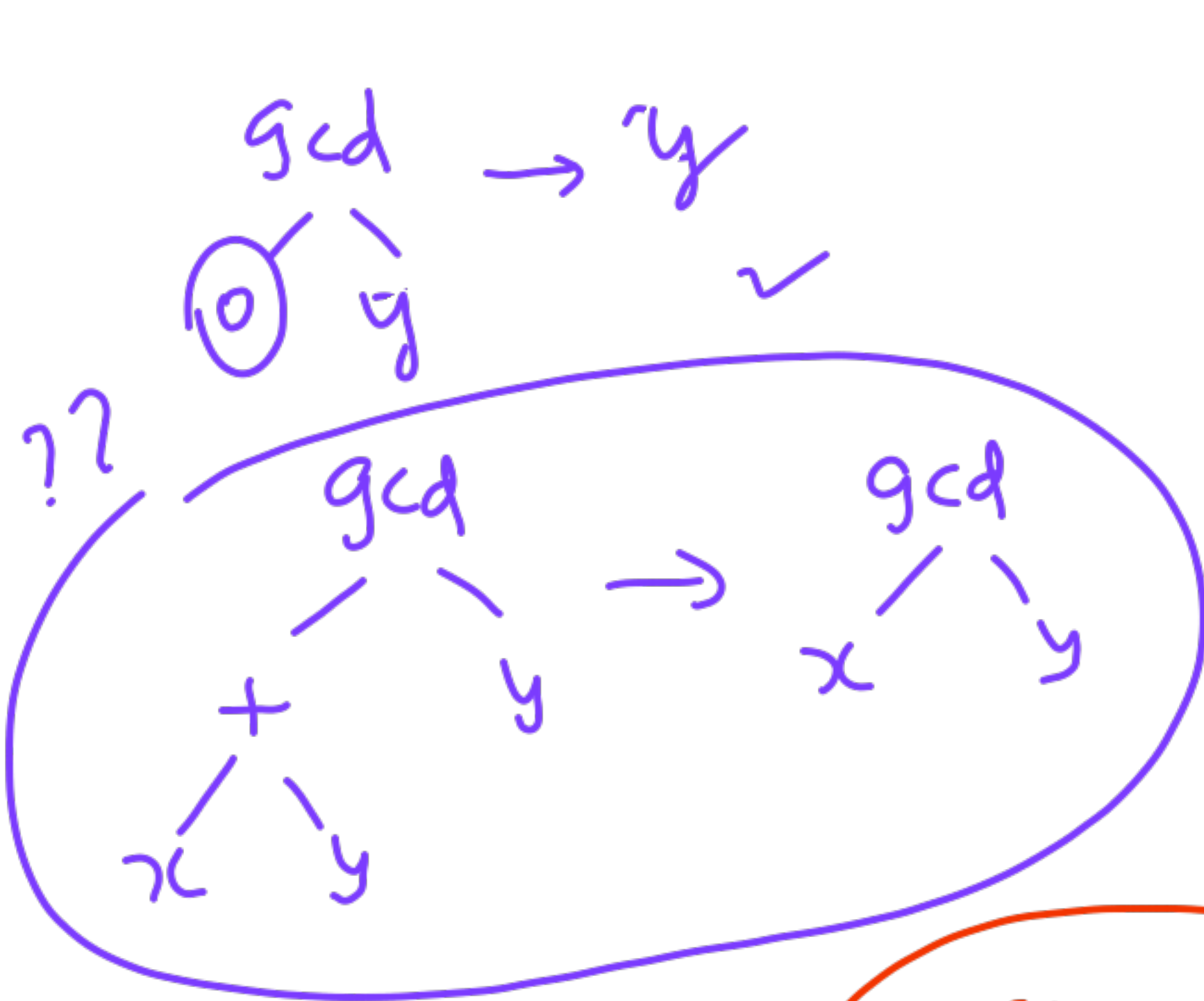
λb
|
 λt
|
 λe
|
 g
|
 b | t | e

"Lazy" eval

Term Rewriting Systems

rules
replace equals by equals
 $l \rightarrow r$





λ -calc $\leftrightarrow$ fnl prog (higher order)

term rew

first order

+ $\rightarrow$ * 10

3 7

Logic Prog.

